

Corrigé — Exercice 1

- 1) $\frac{(3+h)^2 - 9}{h} = 6 + h \rightarrow 6 : f'(3) = 6$. 2) $f'(x) = 6(2x+1)^2$; et $f'(x) = \sqrt{x+1} + \frac{x}{2\sqrt{x+1}} = \frac{3x+2}{2\sqrt{x+1}}$. 3) $f'(x) = 2x - 3, f'(2) = 1, f(2) = -1 : T : y = x - 3$. 4) $f'(x) = \frac{1}{2\sqrt{x}}, f'(1) = \frac{1}{2}, f(1) = 1 : T : y = \frac{1}{2}x + \frac{1}{2}$.
- 5) $f'(x) = \frac{2(x^2+1) - 2x(2x-1)}{(x^2+1)^2} = \frac{-2x^2+2x+2}{(x^2+1)^2}$.
- 6) Tangente horizontale $\iff f'(x) = 2x - 3 = 0 \iff x = \frac{3}{2}$; le point est $\left(\frac{3}{2}; -\frac{5}{4}\right)$.